Pool-ball Detection and Recognition

Manon Yiqin Galian Ng (23375206) – CSU44053

1 Overview

The objective of this project was to locate pool balls in photographs of a table and identify each ball by type and number. I divided the task into detecting candidate balls and recognising their identities, then converted their positions into table coordinates for the final views. Development was iterative: I began with simple approaches and used intermediate images and quantitative errors to decide what to change. AllBalls supplied tuning and colour references; Tables1–6 supported evaluation and refinement.

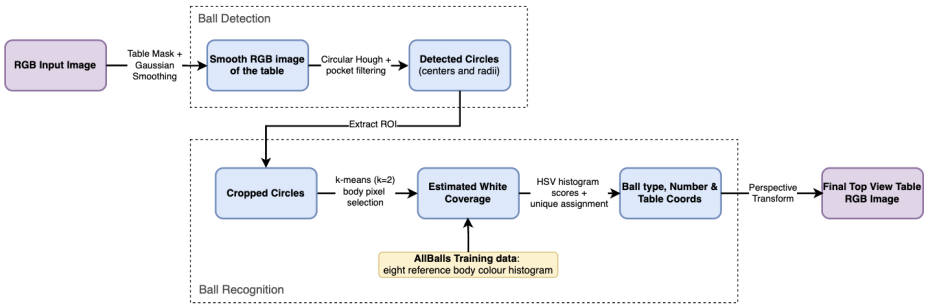


Figure 1: Diagram of the Final Pipeline

2 Detailed Design

2.1 Ball Detection

I used the supplied corners to create a mask of the playing surface and set the background to black. I then converted the image to greyscale and smoothed it with a 5×5 Gaussian filter ($\sigma = 1.5$) to reduce small intensity variations before circular Hough transform.

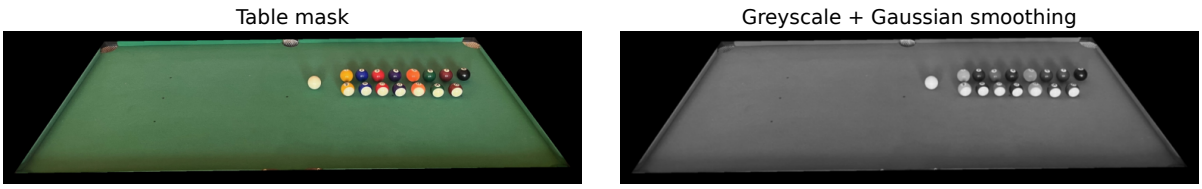


Figure 2: Masking (left) and greyscale smoothing (right) on AllBalls.

I initially varied Hough parameters manually on AllBalls. This was slow and configurations were hard to compare, so I searched and evaluated 480 configurations. The centers were matched to annotations in table coordinates. I ranked settings by most amount of correct detections (as a missed ball cannot be recognised later) then by fewest extras and finally by lowest centre error.



Figure 3: Starting Hough (left), tuned Hough (centre) and pocket filtering (right).

The final selected parameters are as follows: accumulator resolution ratio of 1.5, centre separation 18 pixels, upper Canny threshold of 75 and circle threshold 35, with a fixed radius range of 15–30 pixels. Pockets would often produced circles. So balls with centers within 25 table units of pockets were filted out. This left all 16 balls from 17 AllBalls circles, but there could be risk of removing genuine balls beside pockets.

Finally, each detected candidate was extracted as a square centred on its location, with side length $2r + 1$. A circular mask excluded the square's corner pixels. Keeping the original colours and mask allowed me to compare recognition approaches on the same crop and inspect individual mistakes.

2.2 Ball Type Recognition

I separated recognition into estimating type and body colour. My first type estimator used HLS lightness and saturation to count white, dark and coloured pixels inside each circle. The proportions then determined the type. But, reflections, shadows and imperfect boundaries made these fixed thresholds unreliable: only 43/86 balls were typed correctly.

Next I used k Means ($k = 2$) to cluster the pixels colours. This idea came from the observation that each pool ball has a white (light) section and a coloured one. Then, each cluster was classified using some boundaries that were manually selected. A cluster was considered light if $V \geq 170$ & $S \leq 165$, those with $V \leq 90$ were considered dark and the rest coloured. The fraction of pixels belonging to the light clusters were then used to give the estimated light coverage, noted w. I then used these proportions to estimate the ball type. A ball was classified as White if at least 85% of its masked area was light, or Solid if at least 70% was non-light and the rest as Striped. This improve performace to 76/86.

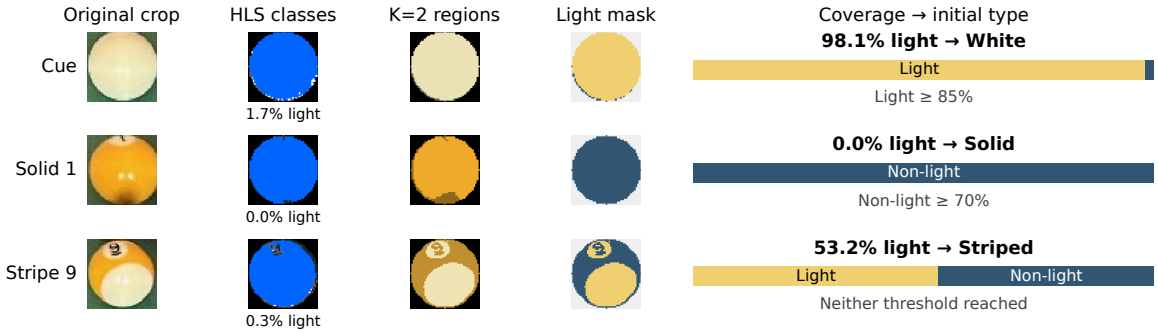


Figure 4: AllBalls cue, solid and stripe examples for HLS and k means

2.3 Ball Colour Recognition

I first applied HSV threshold rules to the mean colour of the largest non-light K-means cluster. But shadows and reflections could distort this single average, so I switched to histograms to preserve the different body colours and their proportions.

I built eight references by grouping labelled AllBalls crops by body colour, pairing solids and stripes. Black is always solid and the cue was excluded. I eroded each circular mask with a 3×3 ellipse and removed pixels with $V \geq 170$, $S \leq 165$. This reduced cloth contamination, shared white stripes, patches and highlights that could make different colours look alike.

Of the histogram resolutions I tested manually, 18 hue bins and eight each for saturation and brightness (1,152 total) gave the highest recognition accuracy. This balanced colour detail against the available pixels. I smoothed neighbouring bins with $[1, 2, 1]/4$ along each axis, wrapping hue. Nearly neutral pixels (BGR range below 15) received hue zero.

I normalised each crop's histogram to sum to one before averaging colour partners, giving both equal influence despite different body-pixel counts. Fewer than 12 body pixels gave an Unknown colour label.

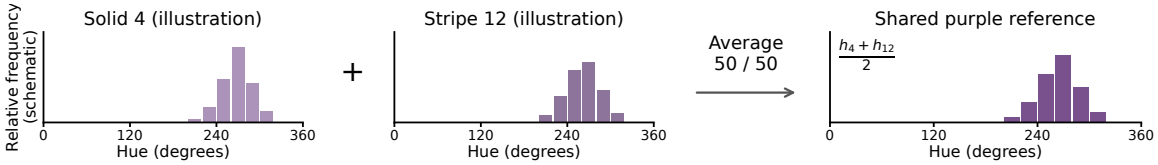


Figure 5: Approximate histogram reference example on purple balls

Finally, candidates were compared with all eight references using one minus Bhattacharyya distance to give similarity $s_{i,c}$ scores. The inital k means approach gave 65/80 and the histogram with the representative-colour rules improved the results to 73/80 with 8 corrections and no regression.

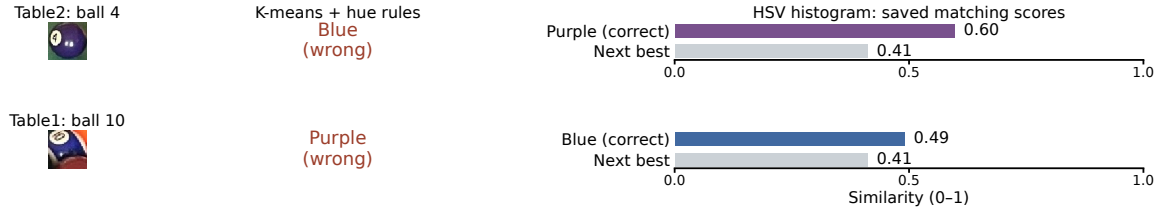


Figure 6: Exported results (csv & ball crops) show two colour corrections.

2.4 Soft Score Approach

A property I wanted to leverage was that each ball has a unique identity. This required scores balls to be assigned scores instead of hard decisions. Therefore, I first replace the single type classification with compatibility scores for Solid, Striped and Cue. I kept the original light-coverage intervals: $[0, 0.30]$ for Solid, $[0.30, 0.85]$ for Striped and $[0.85, 1]$ for White. A type received a score of one inside its interval, with support decreasing gradually outside it:

$$g_t(w) = \exp\left[-\frac{d(w, I_t)^2}{2(0.10)^2}\right]$$

Here $d(w, I_t)$ is distance to the interval, with zero inside it. The Gaussian fall-off shares the RBF form [2], while full support within an interval matches Gaussian combination membership [3]. I chose width 0.10 heuristically; these are compatibility scores, not calibrated probabilities.

For numbered balls, I multiplied histogram colour similarity by type compatibility: $S_{i,n} = s_{i,c(n)} g_{t(n)}(w_i)$. The cue instead used $S_{i,0} = w_i g_{\text{White}}(w_i)$, rewarding high light coverage and adding a penalty below 85%. These scores provided the alternatives needed for the unique assignment stage.

2.5 Unique identity assignment Algorithm

To enforce uniqueness, I used the Hungarian algorithm to maximise the total compatibility score across the table. Each row of the score matrix represented a candidate and each column a possible identity. The algorithm built an assignment while allowing earlier pairings to be rearranged, selecting one identity per candidate and using each identity at most once.

Table2 crop	Score for 4	Score for 12	Assigned identity
Purple solid	**0.597**	0.128	4
Purple stripe	0.639	**0.637**	12

Table 1: Scores example; the two purple crops in Table2

Both crops independently preferred 4. However, assigning the stripe to 12 barely reduced its score, whereas assigning the solid to 12 caused a much larger loss. The selected pairing therefore scored 1.234 overall, compared with 0.767 for the reverse pairing.

This improved correct identities from 73/86 with independent soft decisions to 75/86.

Separate Unknown choices allowed surplus detections or candidates without an eligible identity to remain unassigned. Their near-zero score encouraged best-effort assignment, so weak matches could still receive unused numbers.



Figure 7: Cropped overhead view of AllBalls after perspective correction. AllBalls supplied tuning and colour references.

2.6 Resolving cue-ball ambiguity

Inspection revealed that Hough sometimes found the white patch inside the ball instead of the full ball boundary. Several candidates could then appear White, and uniqueness alone could give the cue to the wrong one. I introduced two refinements.

First, I expanded the radius of originally White crops by one pixel and recomputed their type and colour evidence. The stored centre and original detected radius were unchanged. This exposed pixels outside the first mask and improved correct identities from 75/86 to 79/86.

Second, when several original candidates remained classified as White, I reserved the cue for the largest original radius. A full ball should usually exceed a patch: Table1's radii were 24.15 and 17.55 pixels. This improved performance to 80/86. A more sofiticated approach could account for perspective by comparing each candidate's radius with the expected apparent ball radius at its position on the table.

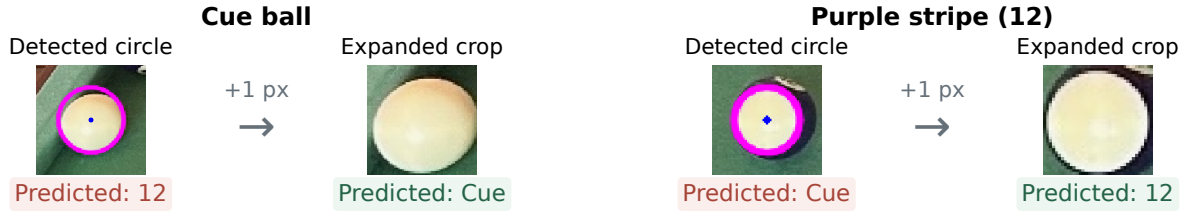


Figure 8: Table3: one-pixel radius expansion corrects the swapped cue and stripe-12 predictions.

3 Results and Analysis

3.1 Evaluation methodology

I evaluated detection and recognition separately to distinguish missed balls from incorrect identities. For detection, I matched predicted centres to annotated positions in the perspective-corrected table, allowing a maximum Euclidean distance of 15 table units. Matching found as many valid pairs as possible, using each detection and annotation only once. Matched pairs counted as true positives (TP), extra detections as false positives (FP), and missed balls as false negatives (FN).

Precision, TP/(TP+FP), measured how many detections corresponded to real balls. Recall, TP/(TP + FN), measured how many annotated balls were found. F1 combined them as 2TP/(2TP + FP + FN). Mean localisation error measured the average centre-to-centre distance of matched pairs.

Recognition accuracy measured the fraction of matched balls given the correct identity. End-to-end accuracy divided correct identities by all annotated balls, so missed balls also counted as failures. Unknown labels counted as incorrect. False detections reduced precision but were excluded from these recognition fractions. Body-colour accuracy excluded the cue.

I reported AllBalls separately because it supplied tuning and colour references. The pooled results cover Tables1–6, which also guided development. They therefore describe performance on images used to refine the system, rather than an independent test.

3.2 Final Metrics

The final system detected 86 of the 90 annotated balls, with two false positives. Precision was **97.73%**, recall **95.56%** and F1 **96.63%**. Mean centre error was 4.91 table units across the 86 matches. Of those matched balls, **80/86** had the correct identity (**93.02%**), giving **80/90**, or **88.89%**, end-to-end accuracy.

The 15-table-unit matching tolerance was a fixed heuristic, approximately one-third of the assumed 43-unit ball diameter. It allowed some uncertainty in detection and annotation while requiring the predicted centre to remain within the ball's nominal radius.

We also evaluated each ball type seperatly, to better observe the strenght and weakness of the proposed system.

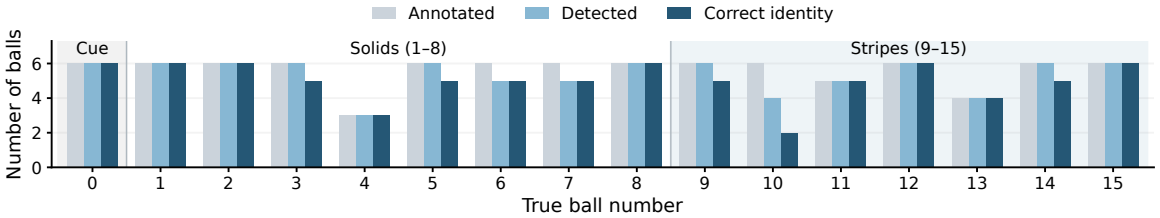


Figure 9: Counts by true identity on Tables1–6 (0 is the cue). Correct identities: solids 41/45, stripes 33/39, cues 6/6; blue stripe 10 was weakest at 2/6.

3.3 Final Table Views

Using the four supplied corners, I calculated a perspective transform that mapped the playing surface to a 1434×683 rectangle, then applied the same transform to the detected ball centres. The warping was done post detection and recognition as to not affect the ball shapes.

Table1: TP 14 | FP 1 | FN 2
IDs 11/16 | mean centre error 7.68

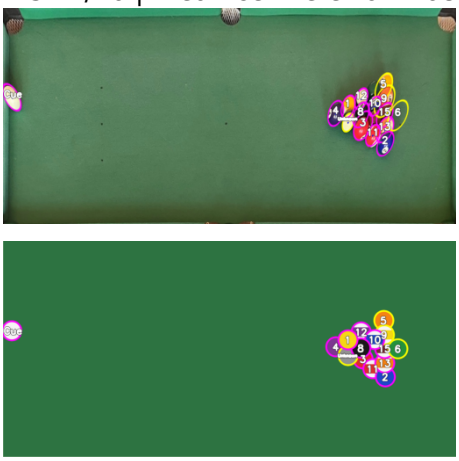


Table2: TP 16 | FP 1 | FN 0
IDs 16/16 | mean centre error 3.58

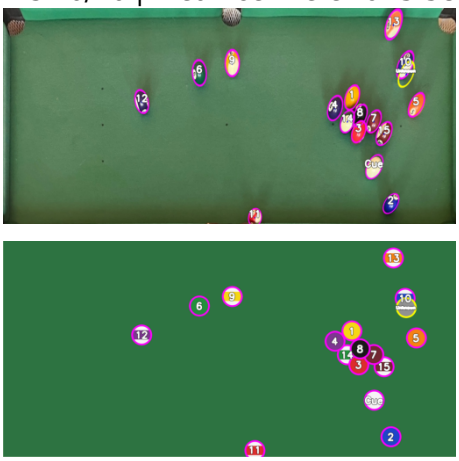


Table3: TP 15 | FP 0 | FN 1
IDs 15/16 | mean centre error 3.12

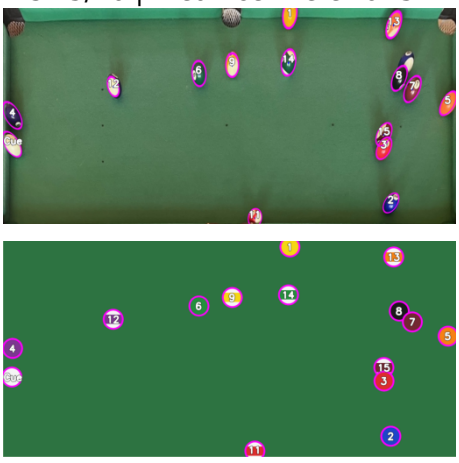
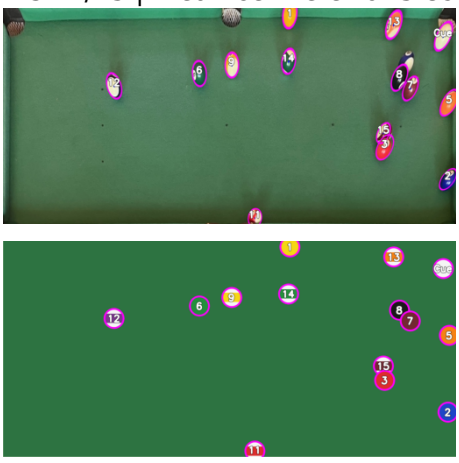


Table4: TP 14 | FP 0 | FN 1
IDs 14/15 | mean centre error 3.89



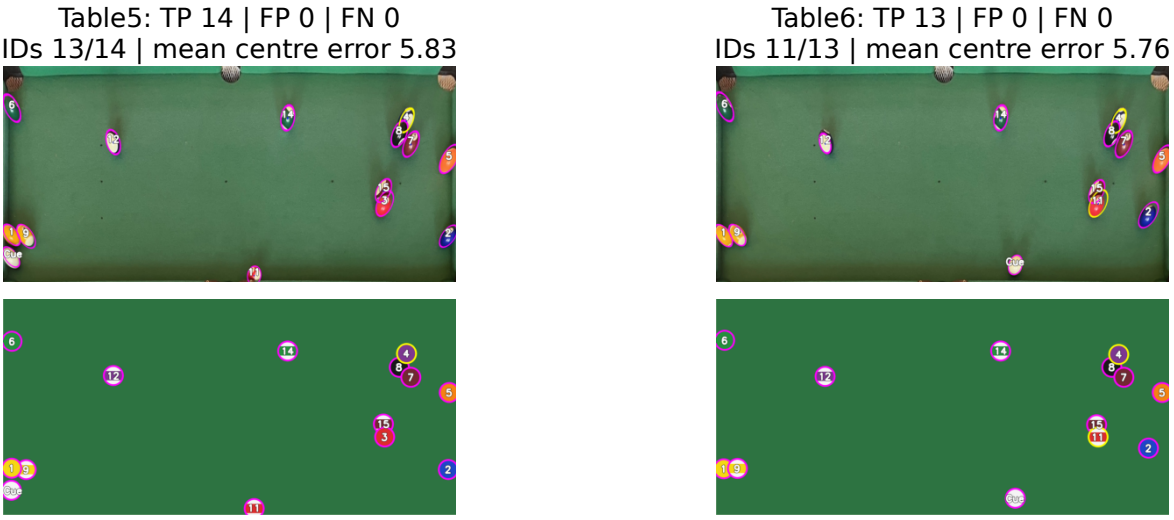


Figure 10: Final outputs for Tables5–6.

The final views pair a numbered rectified photograph with an overhead diagram. The photograph shows the predictions in context, while the diagram uses enlarged markers for readability. Magenta outlines indicate correct identities while yellow indicates wrong identities or false circles, using annotations only for these evaluation colours.

3.4 Observed Strengths and Weaknesses

Table2 was the clearest success: all 16 real balls were correctly identified, while the extra circle remained Unknown. The system could therefore handle some ambiguous appearances and an additional detection without losing correct identities. Tables3 and 4 also identified every detected ball correctly, although both missed ball 10. In these cases, detection was the remaining obstacle to a completely correct result.

Table1 exposed the limits of circle detection in a crowded arrangement. Balls 6 and 7 were missed, a false circle received identity 6, and balls 5 and 9 exchanged labels. The exchanged balls had centre errors of 14.63 and 14.81 table units, close to the 15-unit matching tolerance. They counted as successful detections, but their displaced crops could include neighbouring material or exclude part of the intended ball. A valid centre match therefore did not guarantee useful recognition evidence.

In Table1, the green stripe's crop captured mainly white, leaving too little green for matching. One-pixel padding performed best overall among the sizes tested, but this ball remained Unknown. Table6's blue stripe instead included a dark neighbour and was labelled as a purple solid. Larger crops may recover missing colour, but can also admit neighbours, so increasing padding alone cannot reliably resolve both errors.

Table6's red solid exposed a separate weakness. Its bright red surface passed the light thresholds, inflating white coverage and producing a red-striped label despite the correct body-colour match.

The system performed well overall, but the remaining errors show there is still room to improve detection and recognition.

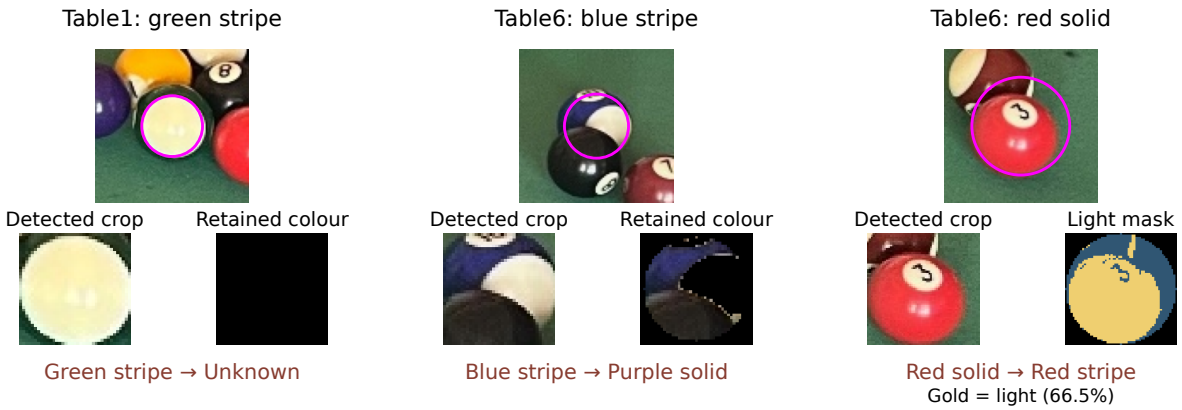


Figure 11: Failure examples: context and detected circles above, crops and evidence below. Gold marks pixels counted as light; black marks excluded pixels.

3.5 Possible Next Steps & Conclusion

The current type classifier uses K-means colour groups and their proportions, without considering their spatial arrangement. Connected-component analysis (CCA) could examine the size, position and connectivity of white and coloured regions, adding evidence to help distinguish solids from stripes.

For detection, guided Hough could focus circle searches on crowded regions using the expected ball size. Watershed could then use the candidate centres as markers to separate touching balls into individual regions. This could help recover missed balls and produce cleaner recognition crops containing less neighbouring material.

Additionally, we could test lighting correction to make white coverage estimates more accurate and consistent across different lighting conditions.

Conclusion. The pipeline performed well for the tested setup. AllBalls provided clear reference examples but limited variety, so the table images were needed to reveal edge cases such as overlapping balls and imperfect crops. These cases helped refine the general approach, while I aimed to limit adjustments for individual images. Additional fully unseen test images would help assess how well these refinements carry over to new arrangements and lighting conditions, with the parameters kept fixed.

References

[1] K. Dawson-Howe, *CSU44053 Computer Vision course material*, Trinity College Dublin.

[2] Scikit-learn Developers, *User Guide*, §1.7.4.3, "Radial basis function (RBF) kernel."

[3] MathWorks, *Fuzzy Logic Toolbox documentation*, "Gaussian combination membership function."

[4] CP-Algorithms, "Hungarian Algorithm."

Code: `https://github.com/galiannm/pool-ball-recogniser`